Supplementary Material

Which reverts count? Reverter populations, attribution rules and observation windows behind a Wikipedia indicator of information quality work

Tables are generated by 03_pipeline/17_supplement_v03.py from the aggregates in 04_output/ of the public repository. Editions marked * were held out for the registered tests. Dump rule: the history dumps' unbounded identity-revert rule. Radius rule: the same hashes with the 15-revision radius of the mwreverts library.

Table S1. Structured review of practice: how studies using a revert-based measure operationalise it

One row per study. "Not stated" means that no statement was found in the methods text read. Evidence level: A, establishing sentence re-verified in the full text; B, taken from a first coder's extract; C, metadata or abstract only. The full table with the establishing sentences, URLs and references is 04_output/practice_review.csv.

short_cite	venue	editions	role	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	evidence_level
Kittur et al., 2007	CHI 2007	en only	conflict indicator	identity + summary matching	none stated	not stated	not stated	excluded	not stated	partial	A
Priedhorsky et al., 2007	GROUP '07	en only	descriptive	identity + summary matching	15 revisions	counted explicitly	counted explicitly	not stated	revision-bounded: 15 revisions	partial	B
Halfaker et al., 2009	WikiSym '09	en only	outcome	identity hash	none stated	not stated	not stated	excluded	unbounded stated	yes: vandalism reverts removed	A
Suh et al., 2009	WikiSym '09	en only	descriptive	summary matching only	none stated	not stated	not stated	not stated	not stated	no	A
Ekstrand & Riedl, 2009	WikiSym '09	en only	descriptive	identity hash	none stated	not stated	not stated	not stated	not stated	no	A
West et al., 2010	EUROSEC '10	en only	ML label	rollback or undo actions only	none stated	excluded	counted explicitly	not stated	not stated	partial	A
Halfaker et al., 2011a	WikiSym '11	en only	exposure	not stated	none stated	not stated	not stated	excluded	not stated	no	A
Halfak	WikiSy	en only	expos	rollback	none	not	not	not	not	no	A

short_cite	venue	editions	role	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	evidence_level
er et al., 2011b	m ’11		ure	or undo actions only	stated	stated	stated	stated	stated		
Lam et al., 2011	WikiSym ’11	en only	outcome	identity + summary matching	none stated	not stated	not stated	not stated	time-bounded: 7 days	no	A
Sumi et al., 2011	IEEE PASSAT/SocialCom 2011	multilingual	conflict indicator	identity hash	none stated	not stated	not stated	excluded	not stated	partial	A
Yasseri et al., 2012	PLoS ONE	en only	conflict indicator	identity hash	none stated	not stated	not stated	excluded	not stated	no	A
Flöck et al., 2012	HT ’12	en only	descriptive	diff-based	20 revisions (also 50, 100)	not stated	not stated	not stated	revision-bounded: 20 revisions	yes: scan range 20/50/100; diff vs identity	A
Halfaker et al., 2013	American Behavioral Scientist	en only	exposure	identity hash	none stated	modelled separately or focal	modelled separately	excluded (definitional)	not stated	no	A
Geiger & Halfaker, 2013	WikiSym ’13	en only	outcome	not stated	none stated	modelled separately or focal	modelled separately	not stated	unbounded stated	no	A
Yasseri et al., 2014	Global Wikipedia (edited book)	multilingual	conflict indicator	identity hash	none stated	not stated	not stated	not stated	not stated	no	A
Tsvetkova et al., 2016	Scientific Reports	multilingual	conflict indicator	identity hash	none stated	excluded	not stated	excluded	time-bounded: 24 hours	yes: window 24-240 h	A
Tsvetkova et al., 2017	PLoS ONE	multilingual	conflict indicator	identity hash	none stated	modelled separately or focal	not stated	excluded	not stated	no	A
Geiger & Halfaker, 2017	PACMHCI (CSCW)	multilingual	conflict indicator	identity hash	mwreverts default (not reported)	modelled separately or focal	not stated	not stated	not stated	partial	A
TeBlunthuis et al., 2018	CHI 2018	Wikia	exposure	not stated	none stated	modelled separately or focal	not stated	not stated	not stated	no	A
DeDeo, 2013	PLoS ONE	en only	conflict indicator	identity hash	none stated	not stated	not stated	excluded	not stated	partial	A
Zhang et al.,	ICWSM 2019	en only	exposure	not stated	none stated	not stated	not stated	excluded	unbounded stated	no	A

short_cite	venue	editions	role	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	evidence_level
2019								(definitional)			
Tran et al., 2020	IEEE S&P 2020	en only	outcome	identity hash	mwreverts default (not reported)	not stated	not stated	excluded	not stated	no	A
Hill & Shaw, 2021	Communication Research	Wikia	outcome	identity hash	mwreverts default (not reported)	not stated	not stated	not stated	not stated	no	A
Chhabra et al., 2020	OpenSym 2020	en only	conflict indicator	identity hash	none stated	not stated	not stated	counted explicitly	not stated	no	A
TeBlunthuis et al., 2021	PACMHCI (CSCW)	multilingual	outcome	identity hash	none stated	not stated	not stated	excluded	time-bounded: 48 hours	no	A
Ruprechter et al., 2021	Scientific Reports	multilingual	descriptive	dump precomputed field	none stated	counted explicitly	not stated	not stated	not stated	no	A
Wikimedia Foundation, 2021	Meta-Wiki report	single non-English edition (pt)	outcome	not stated	none stated	not stated	not stated	not stated	time-bounded: 48 hours	no	A
Tran et al., 2022	PACMHCI (CSCW)	multilingual	outcome	not stated	none stated	not stated	not stated	not stated	unbounded stated	no	A
Wang & Zhu, 2022	FAccT ’22	multilingual	outcome	not stated	none stated	not stated	not stated	excluded (definitional)	not stated	no	A
Champion & Hill, 2023	PACMHCI (CSCW)	en only	outcome	not stated	10 contributions forward	not stated	not stated	not stated	revision-bounded: 10 contributions	no	A
Warncke-Wang et al., 2023	PACMHCI (CSCW)	multilingual	outcome	not stated	none stated	not stated	not stated	not stated	time-bounded: 48 hours	no	A
Trokhymovych et al., 2023	KDD ’23	multilingual	ML label	dump precomputed field	none stated	not stated	not stated	not stated	not stated	no	A
Franzmeyer et al., 2024	Findings of ACL 2024	en only	ML label	not stated	none stated	not stated	not stated	excluded (definitional)	not stated	no	A
Formisano et al., 2026	Humanities and Social Sciences Commu	en only	outcome	dump precomputed field	none stated	not stated	not stated	not stated	not stated	no	A

short_cite	venue	editions	role	detection	radius_or_rule	bot_reverts	tool_reverts	self_reverts	window	sensitivity_over_revert_choices	evidence_level
	nications										

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Table S2. Composition of first reverts of newcomer first edits (dump rule, full cohorts)

First edits 2019 onward, window 24 h. Shares of first reverts with 95 percent Wilson intervals.

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	2,189,284	0.37	0.122 [0.122, 0.123]	0.448 [0.447, 0.449]	0.364 [0.363, 0.365]	0.023 [0.022, 0.023]	0.044 [0.043, 0.044]
Italian*	135,400	0.342	0.000 [0.000, 0.000]	0.525 [0.520, 0.529]	0.407 [0.402, 0.411]	0.024 [0.023, 0.026]	0.044 [0.042, 0.046]
Korean	47,633	0.216	0.003 [0.002, 0.004]	0.516 [0.506, 0.525]	0.289 [0.280, 0.298]	0.096 [0.091, 0.102]	0.096 [0.091, 0.102]
German	236,903	0.256	0.001 [0.001, 0.002]	0.410 [0.406, 0.414]	0.545 [0.541, 0.549]	0.017 [0.016, 0.018]	0.027 [0.026, 0.028]
Polish*	67,755	0.222	0.001 [0.001, 0.002]	0.315 [0.307, 0.322]	0.616 [0.608, 0.624]	0.030 [0.028, 0.033]	0.038 [0.035, 0.041]
French	346,938	0.243	0.042 [0.041, 0.043]	0.255 [0.252, 0.258]	0.643 [0.640, 0.647]	0.012 [0.012, 0.013]	0.047 [0.046, 0.049]
Japanese*	223,045	0.14	0.001 [0.001, 0.001]	0.247 [0.242, 0.252]	0.558 [0.552, 0.563]	0.080 [0.077, 0.083]	0.114 [0.111, 0.118]

First edits 2019 onward, window unbounded. Shares of first reverts with 95 percent Wilson intervals.

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	2,189,284	0.412	0.111 [0.111, 0.112]	0.432 [0.431, 0.433]	0.383 [0.382, 0.384]	0.034 [0.033, 0.034]	0.040 [0.040, 0.041]
Italian*	135,400	0.369	0.001 [0.001, 0.001]	0.503 [0.498, 0.507]	0.421 [0.417, 0.425]	0.034 [0.033, 0.036]	0.041 [0.040, 0.043]
Korean	47,633	0.279	0.006 [0.005, 0.007]	0.438 [0.429, 0.446]	0.336 [0.328, 0.344]	0.142 [0.137, 0.148]	0.078 [0.074, 0.083]

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
German	236,903	0.294	0.003 [0.002, 0.003]	0.369 [0.365, 0.373]	0.586 [0.582, 0.590]	0.017 [0.016, 0.018]	0.025 [0.024, 0.026]
Polish*	67,755	0.303	0.002 [0.001, 0.003]	0.248 [0.242, 0.254]	0.690 [0.684, 0.696]	0.031 [0.029, 0.034]	0.029 [0.026, 0.031]
French	346,938	0.27	0.039 [0.038, 0.040]	0.241 [0.238, 0.243]	0.654 [0.651, 0.657]	0.023 [0.022, 0.023]	0.044 [0.042, 0.045]
Japanese*	223,045	0.172	0.003 [0.003, 0.004]	0.209 [0.205, 0.213]	0.581 [0.576, 0.586]	0.109 [0.106, 0.113]	0.097 [0.094, 0.100]

First edits 2021 onward, window 24 h. Shares of first reverts with 95 percent Wilson intervals.

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	1,453,656	0.361	0.125 [0.124, 0.126]	0.524 [0.522, 0.525]	0.285 [0.284, 0.287]	0.022 [0.022, 0.022]	0.044 [0.043, 0.044]
Italian*	88,005	0.334	0.000 [0.000, 0.000]	0.546 [0.540, 0.551]	0.390 [0.385, 0.396]	0.020 [0.019, 0.022]	0.044 [0.041, 0.046]
Korean	32,751	0.239	0.001 [0.001, 0.002]	0.554 [0.543, 0.565]	0.262 [0.252, 0.272]	0.094 [0.088, 0.100]	0.090 [0.084, 0.096]
German	159,216	0.248	0.001 [0.001, 0.002]	0.394 [0.389, 0.398]	0.560 [0.555, 0.564]	0.018 [0.017, 0.019]	0.028 [0.026, 0.029]
Polish*	45,382	0.227	0.002 [0.001, 0.003]	0.303 [0.294, 0.312]	0.624 [0.614, 0.633]	0.035 [0.031, 0.039]	0.037 [0.033, 0.040]
French	229,556	0.232	0.020 [0.019, 0.022]	0.261 [0.257, 0.265]	0.656 [0.652, 0.660]	0.011 [0.010, 0.012]	0.051 [0.049, 0.053]
Japanese*	148,508	0.138	0.001 [0.001, 0.002]	0.300 [0.294, 0.306]	0.519 [0.512, 0.526]	0.069 [0.066, 0.073]	0.110 [0.106, 0.114]

First edits 2021 onward, window unbounded. Shares of first reverts with 95 percent Wilson intervals.

Edition	First edits	p reverted	Bot	One-click tool	Other registered, manual	Unregistered	Self
English	1,453,656	0.402	0.114 [0.113, 0.114]	0.508 [0.507, 0.510]	0.305 [0.304, 0.306]	0.033 [0.032, 0.033]	0.040 [0.040, 0.041]
Italian*	88,005	0.361	0.001 [0.000, 0.001]	0.522 [0.517, 0.528]	0.406 [0.401, 0.412]	0.030 [0.028, 0.031]	0.041 [0.039, 0.043]
Korean	32,751	0.303	0.004 [0.003, 0.005]	0.477 [0.467, 0.487]	0.310 [0.301, 0.319]	0.135 [0.129, 0.142]	0.074 [0.069, 0.080]
German	159,216	0.287	0.003 [0.002, 0.003]	0.353 [0.349, 0.358]	0.600 [0.596, 0.605]	0.018 [0.017, 0.020]	0.025 [0.024, 0.027]
Polish*	45,382	0.316	0.002 [0.001, 0.003]	0.234 [0.227, 0.241]	0.704 [0.697, 0.712]	0.033 [0.030, 0.036]	0.027 [0.025, 0.030]
French	229,556	0.26	0.019 [0.018, 0.021]	0.246 [0.242, 0.249]	0.667 [0.663, 0.671]	0.020 [0.019, 0.022]	0.047 [0.045, 0.049]
Japanese*	148,508	0.17	0.004 [0.003, 0.005]	0.254 [0.249, 0.259]	0.553 [0.547, 0.559]	0.096 [0.093, 0.100]	0.093 [0.089, 0.097]

Table S3. Change in the revert rate when a class of first reverts is recoded as not reverted (percentage points; identity: share times rate)

First edits 2019 onward, 24-hour window.

Edition	Bots	One-click tools	Self-reverts	Unregistered reverters	Undo-button reverts (if counted as a tool)	Undo share of manual reverts
English	4.53	16.56	1.62	0.84	11.24	0.707
Italian*	0.01	17.93	1.49	0.83	10.17	0.627
Korean	0.07	11.13	2.07	2.07	5.87	0.565
German	0.04	10.5	0.69	0.43	7.31	0.485
Polish*	0.03	6.99	0.83	0.67	7.54	0.497
French	1.02	6.2	1.15	0.3	11.69	0.684
Japanese*	0.01	3.45	1.6	1.11	5.69	0.542

First edits 2021 onward, 24-hour window.

Edition	Bots	One-click tools	Self-reverts	Unregistered reverters	Undo-button reverts (if counted as a tool)	Undo share of manual reverts
English	4.51	18.9	1.58	0.79	8.49	0.67
Italian*	0	18.21	1.46	0.67	9.76	0.645
Korean	0.03	13.25	2.15	2.24	5.97	0.561
German	0.03	9.76	0.68	0.45	7.19	0.479
Polish*	0.04	6.87	0.83	0.79	8.37	0.531
French	0.48	6.07	1.19	0.26	11.81	0.707
Japanese*	0.02	4.15	1.52	0.96	5.46	0.565

Table S4. Follow-up manual reverts by class of first reverter, and the window ratio R, under two attribution rules (page-history samples)

Dump rule, window 24 h. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.100	0.114	0.122	0.089	0.111	0.120	0.145
Manual, by another editor	0.149	0.175	0.239	0.091	0.098	0.151	0.213
Bot	0.142	0.000	0.129	0.065	0.056	0.237	0.000
One-click tool	0.089	0.114	0.122	0.089	0.112	0.101	0.146
Other registered, manual	0.146	0.171	0.241	0.090	0.095	0.151	0.206
Unregistered	0.204	0.239	0.233	0.135	0.162	0.175	0.263
Self	0.134	0.134	0.276	0.149	0.159	0.114	0.147

Dump rule, window 7 d. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.147	0.174	0.210	0.117	0.154	0.168	0.180
Manual, by another editor	0.209	0.234	0.312	0.119	0.140	0.205	0.262
Bot	0.194	0.286	0.250	0.083	0.048	0.295	0.222
One-click tool	0.134	0.174	0.210	0.117	0.155	0.148	0.180
Other registered, manual	0.207	0.231	0.312	0.118	0.137	0.204	0.256
Unregistered	0.236	0.279	0.313	0.159	0.202	0.223	0.299

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Self	0.178	0.199	0.319	0.165	0.179	0.147	0.164

Dump rule, window 180 d. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.211	0.267	0.356	0.186	0.256	0.264	0.251
Manual, by another editor	0.277	0.312	0.397	0.174	0.198	0.283	0.315
Bot	0.259	0.455	0.288	0.148	0.086	0.368	0.136
One-click tool	0.199	0.267	0.357	0.186	0.257	0.248	0.252
Other registered, manual	0.277	0.310	0.388	0.173	0.195	0.284	0.313
Unregistered	0.270	0.325	0.420	0.209	0.284	0.263	0.326
Self	0.222	0.277	0.400	0.231	0.254	0.215	0.209

Dump rule, window unbounded. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.217	0.279	0.385	0.198	0.280	0.279	0.258
Manual, by another editor	0.282	0.318	0.419	0.187	0.217	0.293	0.320
Bot	0.266	0.357	0.325	0.159	0.108	0.382	0.160
One-click tool	0.204	0.279	0.386	0.199	0.281	0.263	0.259
Other registered, manual	0.283	0.317	0.408	0.186	0.213	0.294	0.318
Unregistered	0.274	0.326	0.447	0.230	0.286	0.279	0.332
Self	0.227	0.281	0.417	0.240	0.264	0.232	0.212

Dump rule, unbounded: follow-up after the first edit's content was reinstated (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.015	0.023	0.127	0.020	0.033	0.012	0.044
Manual, by another editor	0.023	0.031	0.092	0.024	0.020	0.022	0.060

Dump rule, unbounded: follow-up immediately, restoring a still earlier state (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.015	0.030	0.046	0.025	0.046	0.020	0.018
Manual, by another editor	0.020	0.024	0.044	0.007	0.012	0.017	0.023

Dump rule, unbounded: follow-up after other intervening edits (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.187	0.227	0.212	0.153	0.200	0.247	0.196
Manual, by another editor	0.239	0.263	0.283	0.156	0.185	0.253	0.238

Radius rule, window 24 h. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.017	0.033	0.048	0.028	0.048	0.027	0.063
Manual, by another editor	0.020	0.027	0.051	0.008	0.012	0.016	0.030
Bot	0.022	0.000	0.069	0.032	0.056	0.059	0.000
One-click tool	0.016	0.033	0.048	0.028	0.048	0.021	0.063
Other registered, manual	0.019	0.025	0.049	0.008	0.011	0.015	0.031
Unregistered	0.043	0.063	0.054	0.009	0.033	0.036	0.025
Self	0.036	0.046	0.118	0.067	0.081	0.035	0.071

Radius rule, window 7 d. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.022	0.038	0.059	0.030	0.052	0.030	0.065
Manual, by another editor	0.023	0.031	0.056	0.008	0.015	0.019	0.035
Bot	0.028	0.000	0.067	0.028	0.048	0.063	0.000
One-click tool	0.021	0.038	0.059	0.030	0.052	0.025	0.066
Other registered, manual	0.022	0.029	0.053	0.008	0.014	0.018	0.036
Unregistered	0.035	0.067	0.064	0.012	0.037	0.040	0.030
Self	0.045	0.059	0.127	0.068	0.081	0.046	0.072

Radius rule, window 180 d. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.027	0.044	0.072	0.033	0.060	0.033	0.071
Manual, by another editor	0.027	0.035	0.063	0.011	0.016	0.022	0.040
Bot	0.033	0.000	0.057	0.049	0.029	0.065	0.000
One-click tool	0.025	0.044	0.073	0.033	0.060	0.028	0.071
Other registered, manual	0.026	0.033	0.060	0.010	0.015	0.022	0.041
Unregistered	0.035	0.061	0.071	0.023	0.048	0.035	0.031
Self	0.051	0.067	0.137	0.073	0.085	0.049	0.075

Radius rule, window unbounded. Share of first edits first reverted by the class shown that were later reverted manually by another editor.

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.027	0.044	0.074	0.033	0.060	0.034	0.070
Manual, by another editor	0.027	0.036	0.064	0.011	0.017	0.022	0.040
Bot	0.034	0.000	0.079	0.048	0.027	0.066	0.000
One-click tool	0.025	0.044	0.074	0.033	0.061	0.029	0.071
Other registered, manual	0.026	0.034	0.061	0.011	0.016	0.022	0.042
Unregistered	0.035	0.060	0.072	0.023	0.047	0.034	0.032
Self	0.051	0.067	0.136	0.075	0.087	0.049	0.075

Radius rule, unbounded: follow-up after the first edit's content was reinstated (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.001	0.002	0.010	0.002	0.004	0.001	0.004
Manual, by another editor	0.001	0.002	0.008	0.001	0.001	0.000	0.003

Radius rule, unbounded: follow-up immediately, restoring a still earlier state (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.015	0.029	0.040	0.025	0.045	0.020	0.018
Manual, by another editor	0.018	0.022	0.038	0.007	0.011	0.017	0.021

Radius rule, unbounded: follow-up after other intervening edits (share of first edits first reverted by the class).

first_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Bot or one-click tool	0.011	0.014	0.025	0.007	0.011	0.013	0.049
Manual, by another editor	0.008	0.012	0.017	0.003	0.005	0.005	0.017

Window ratio R (replace effect of excluding bots and tools at the longer window divided by the effect within 24 hours; 2,000 bootstrap resamples of editors).

key	English	Italian*	Korean	German	Polish*	French	Japanese*
Dump rule, 180 d / 24 h	0.926 [0.919, 0.934]	0.857 [0.848, 0.865]	0.809 [0.793, 0.825]	0.931 [0.921, 0.941]	0.905 [0.890, 0.920]	0.875 [0.861, 0.889]	0.926 [0.906, 0.945]
Dump rule, unbounded / 24 h	0.922 [0.915, 0.930]	0.843 [0.834, 0.852]	0.775 [0.758, 0.792]	0.917 [0.907, 0.928]	0.877 [0.860, 0.892]	0.858 [0.844, 0.871]	0.919 [0.898, 0.939]
Radius rule, 180 d / 24 h	1.043 [1.038, 1.048]	1.024 [1.020, 1.028]	1.074 [1.064, 1.085]	1.036 [1.030, 1.042]	1.067 [1.058, 1.077]	1.039 [1.032, 1.046]	1.046 [1.034, 1.058]
Radius rule, unbounded / 24 h	1.045 [1.041, 1.050]	1.024 [1.020, 1.028]	1.074 [1.065, 1.085]	1.036 [1.030, 1.042]	1.067 [1.057, 1.077]	1.039 [1.031, 1.046]	1.048 [1.036, 1.061]

Table S5. Retention (an edit on days 2 to 60) by who made the first revert within 24 hours

Cells: share retained; odds ratio against newcomers not reverted within 24 hours with 95 percent Woolf interval (0.5 added to each cell); number of newcomers. Classes with fewer than 30 newcomers are omitted.

first_revert_class	English	Italian*	Korean	German	Polish*	French	Japanese*
Any revert	0.097; OR 0.43 [0.43, 0.43]; n = 809,720	0.090; OR 0.40 [0.38, 0.41]; n = 46,260	0.079; OR 0.32 [0.29, 0.34]; n = 10,279	0.107; OR 0.51 [0.50, 0.53]; n = 60,665	0.113; OR 0.54 [0.51, 0.57]; n = 15,033	0.089; OR 0.47 [0.46, 0.49]; n = 84,351	0.116; OR 0.36 [0.35, 0.37]; n = 31,127
Bot	0.069; OR 0.30 [0.29, 0.31]; n = 99,135	nan	0.194; OR 0.94 [0.40, 2.22]; n = 31	0.081; OR 0.40 [0.19, 0.85]; n = 86	nan	0.041; OR 0.21 [0.18, 0.25]; n = 3,552	nan
One-click tool	0.086; OR 0.38 [0.37, 0.38]; n = 362,541	0.072; OR 0.31 [0.29, 0.33]; n = 24,278	0.035; OR 0.13 [0.12, 0.16]; n = 5,301	0.062; OR 0.29 [0.27, 0.30]; n = 24,876	0.067; OR 0.30 [0.27, 0.34]; n = 4,733	0.052; OR 0.27 [0.25, 0.28]; n = 21,513	0.037; OR 0.10 [0.09, 0.12]; n = 7,690
Other registered, manual	0.114; OR 0.52 [0.51, 0.52]; n = 294,376	0.108; OR 0.48 [0.46, 0.51]; n = 18,823	0.118; OR 0.50 [0.44, 0.56]; n = 2,971	0.142; OR 0.71 [0.68, 0.73]; n = 33,055	0.135; OR 0.66 [0.62, 0.70]; n = 9,263	0.104; OR 0.57 [0.55, 0.58]; n = 54,268	0.129; OR 0.41 [0.39, 0.43]; n = 17,362
Unregistered	0.106; OR 0.48 [0.45, 0.50]; n = 18,309	0.121; OR 0.55 [0.46, 0.66]; n = 1,130	0.122; OR 0.52 [0.43, 0.63]; n = 988	0.084; OR 0.40 [0.32, 0.50]; n = 1,007	0.132; OR 0.65 [0.50, 0.85]; n = 454	0.082; OR 0.44 [0.35, 0.55]; n = 1,044	0.152; OR 0.49 [0.44, 0.55]; n = 2,485
Self	0.139; OR 0.65 [0.63, 0.67]; n = 35,359	0.128; OR 0.59 [0.51, 0.67]; n = 2,017	0.153; OR 0.67 [0.56, 0.79]; n = 988	0.113; OR 0.55 [0.47, 0.64]; n = 1,641	0.124; OR 0.61 [0.47, 0.78]; n = 564	0.119; OR 0.66 [0.60, 0.73]; n = 3,974	0.199; OR 0.68 [0.63, 0.74]; n = 3,562

Share retained among newcomers not reverted within 24 hours:

English	Italian*	Korean	German	Polish*	French	Japanese*
0.199	0.201	0.213	0.189	0.191	0.171	0.266

Table S6. Tag coverage: radius-rule reverts whose reverted first edit carries no mw-reverted tag, first edits from October 2020

Patrolling regime: English, Korean, Japanese none; German, Polish FlaggedRevs; Italian, French recent-changes patrolling. Cells: untagged share with 95 percent Wilson interval; number of reverts.

reverter_class	English	Italian*	Korean	German	Polish*	French	Japanese*
All reverters	0.003 [0.002, 0.004]; n = 16,611	0.096 [0.091, 0.101]; n = 14,935	0.004 [0.003, 0.005]; n = 9,485	0.024 [0.022, 0.027]; n = 12,246	0.039 [0.036, 0.043]; n = 13,282	0.070 [0.065, 0.075]; n = 10,914	0.010 [0.008, 0.013]; n = 7,039
Bot	0.000 [0.000, 0.003]; n = 2,012	0.000 [0.000, 0.390]; n = 6	0.000 [0.000, 0.071]; n = 50	0.043 [0.012, 0.145]; n = 46	0.161 [0.071, 0.326]; n = 31	0.004 [0.001, 0.024]; n = 231	0.048 [0.008, 0.227]; n = 21

reverter_class	English	Italian*	Korean	German	Polish*	French	Japanese*
One-click tool	0.002 [0.001, 0.003]; n = 8,433	0.006 [0.005, 0.008]; n = 7,952	0.002 [0.001, 0.004]; n = 4,809	0.022 [0.018, 0.027]; n = 4,413	0.052 [0.045, 0.061]; n = 3,112	0.003 [0.002, 0.006]; n = 2,682	0.015 [0.010, 0.022]; n = 1,813
Other registered, manual	0.003 [0.002, 0.005]; n = 4,942	0.106 [0.099, 0.114]; n = 5,964	0.004 [0.002, 0.007]; n = 2,793	0.016 [0.013, 0.019]; n = 7,305	0.020 [0.017, 0.023]; n = 9,363	0.031 [0.027, 0.035]; n = 7,274	0.006 [0.004, 0.009]; n = 3,884
Unregistered	0.002 [0.000, 0.010]; n = 551	0.786 [0.745, 0.822]; n = 434	0.002 [0.000, 0.007]; n = 1,107	0.235 [0.179, 0.301]; n = 183	0.286 [0.245, 0.332]; n = 412	0.715 [0.653, 0.770]; n = 228	0.000 [0.000, 0.006]; n = 624
Self	0.022 [0.014, 0.036]; n = 673	0.708 [0.670, 0.744]; n = 579	0.022 [0.014, 0.035]; n = 726	0.127 [0.094, 0.170]; n = 299	0.135 [0.103, 0.174]; n = 364	0.733 [0.693, 0.770]; n = 499	0.027 [0.018, 0.042]; n = 697

Table S7. Registered tests in the held-out editions, as produced by the registered code

Bootstrap replicates in the registered code: 2,000 for H1 and H2, 500 for H4 (the manuscript reports H4 intervals recomputed with 2,000 replicates, below). Holm adjustment across the three tests per edition does not change any decision.

	Japanese*	Polish*	Italian*
n	60000.000	59997.000	59999.000
delta24_pp	2.972	6.217	15.900
deltainf_pp	2.730	5.450	13.409
D_pp	0.242	0.767	2.492
D_ci90_lo	0.188	0.680	2.362
D_ci95_lo	0.180	0.660	2.338
D_ci95_hi	0.307	0.872	2.650
R	0.919	0.877	0.843
R_ci_lo	0.898	0.861	0.834
R_ci_hi	0.939	0.892	0.852
h2_n	60000.000	59997.000	59999.000
h2_delta_strict_pp	3.477	6.995	17.955
h2_delta_replace_pp	2.972	6.217	15.900
h2_S	0.145	0.111	0.114
h2_S_ci95_hi_onesided	0.157	0.119	0.119
h2_S_ci_lo	0.131	0.102	0.109
h2_S_ci_hi	0.160	0.120	0.121
h3_n	44211.000	44041.000	43006.000
h3_agree_d2_d3r15	0.986	0.979	0.954
h3_agree_d1_d2	0.981	0.977	0.951
h3_n_d1only	538.000	876.000	1997.000
h3_d1only_beyond_radius	0.424	0.094	0.073
h4_or_ratio	1.836	1.425	1.783
h4_or_ratio_ci_lo	1.791	1.375	1.720
h4_or_ratio_ci_hi	1.878	1.494	1.841
h4_boot	500.000	500.000	500.000

H4 recomputed with 2,000 paired bootstrap replicates (all editions).

Edition	max/min odds ratio	95% interval	one-sided 95% lower bound
English	1.785	[1.768, 1.801]	1.771
Italian*	1.783	[1.722, 1.846]	1.732
Korean	3.008	[2.801, 3.221]	2.84
German	1.629	[1.592, 1.665]	1.599
Polish*	1.425	[1.368, 1.491]	1.378
French	1.512	[1.485, 1.538]	1.489
Japanese*	1.836	[1.791, 1.887]	1.798

Table S8. Change tags on reverting revisions by edition and year (all namespaces)

mw-rollback

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	1122426	895626	932095	883106	750250	774662	589221	606866	445436
Italian*	119804	124817	147566	159560	156810	127405	103429	90733	44813
Korean	20961	14881	28911	22528	31437	27802	29237	28939	11935
German	193873	189579	154538	142501	117173	120366	97322	69006	44425
Polish*	33863	33984	42665	37281	36801	33087	34604	22962	12179
French	82033	86959	68743	76641	67687	60521	59402	47767	33146
Japanese*	9121	11759	18030	27752	24273	26174	19051	14461	8951

huggle

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	345176	262033	162423	1631	2409	151939	132999	104257	47488
Italian*	0	0	0	0	0	0	0	0	0
Korean	0	0	0	0	0	0	0	0	0
German	0	0	0	0	0	0	0	0	0
Polish*	0	0	0	0	0	0	0	0	0
French	5016	4433	836	1111	1774	4027	1868	1510	280
Japanese*	0	0	0	0	0	0	0	0	0

twinkle

	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	0	0	207362	602187	543831	514898	541969	538128	322779
Italian*	0	0	0	0	0	0	0	0	0
Korean	0	0	0	0	0	0	0	0	0
German	0	0	0	0	0	0	0	0	0
Polish*	0	0	0	0	0	0	0	0	0
French	0	0	0	0	0	0	0	0	0
Japanese*	0	0	0	0	0	0	0	0	0

mw-undo

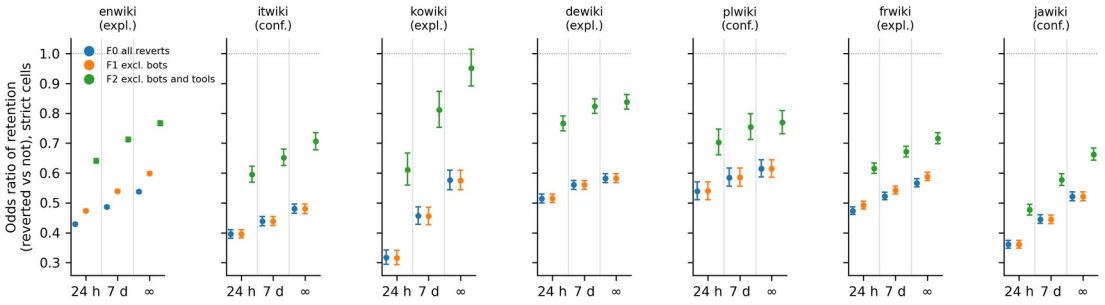
	2018	2019	2020	2021	2022	2023	2024	2025	2026
English	1494822	1529089	1540633	1552495	1435631	1340784	1362885	1285981	803483
Italian*	95578	117098	116429	114948	97846	89979	84370	73064	45823
Korean	30946	29764	57931	45761	26982	21147	28007	21249	11311
German	109547	96185	98655	97514	81234	79920	75943	69172	44466
Polish*	24450	26055	33152	46991	49128	31284	32403	22045	13004
French	161866	154328	168055	151144	134478	137671	119977	118490	84144

	2018	2019	2020	2021	2022	2023	2024	2025	2026
Japanese*	74816	95772	95019	93695	94968	67232	50025	44876	31462

Table S9. Cohort variants: bot-or-tool share of first reverts within 24 hours

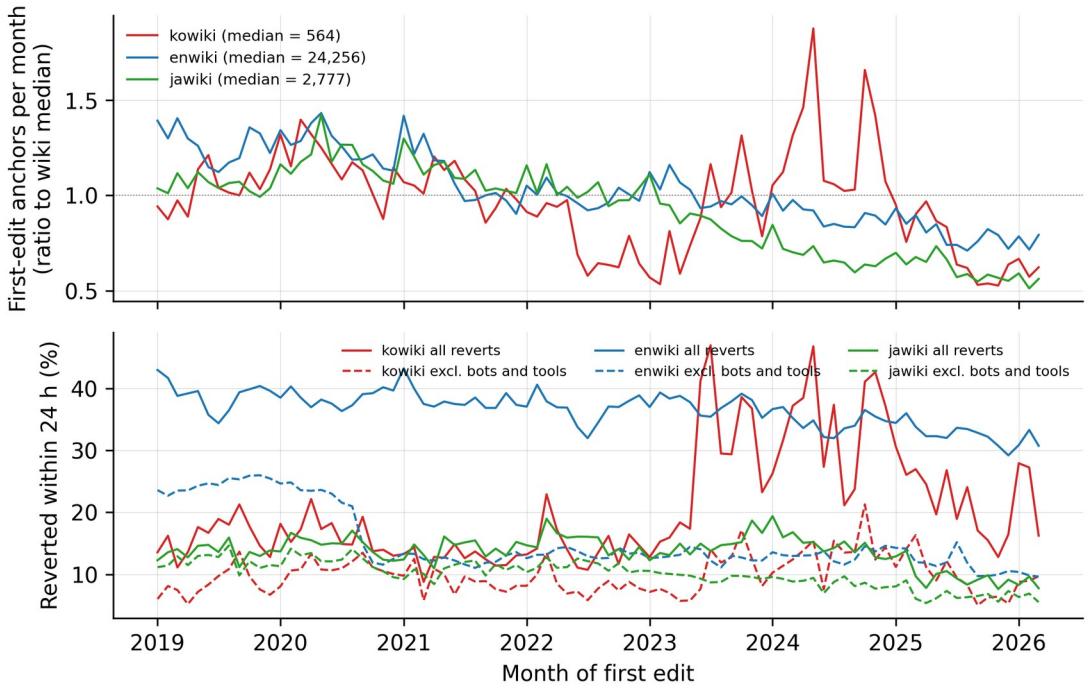
cohort	English	Italian*	Korean	German	Polish*	French	Japanese*
all registered first edits 2019+	0.565	0.523	0.511	0.398	0.317	0.292	0.259
self-created first edits 2019+ (primary)	0.570	0.525	0.519	0.411	0.316	0.297	0.248
self-created, anchors 2021+ (all tags available)	0.649	0.546	0.555	0.395	0.305	0.282	0.301
self-created, secondary anchors (edits 2-5 within 91 d)	0.564	0.517	0.512	0.409	0.335	0.315	0.280

Figure S1. Odds ratio of retention in the nine registered cells, by edition



Reverted versus non-reverted first edits under three reverter filters (all reverts; without bots; without bots and one-click tools, strict recoding) and three windows. The movement across cells is compositional (see main text).

Figure S2. Monthly newcomer first edits and 24-hour revert rates in Korean, English and Japanese, 2019 to March 2026



Upper panel: first edits per month relative to the edition's median. Lower panel: share reverted within 24 hours, all reverts (solid) and without bots and one-click tools (dashed). The drop of the dashed English line in September 2020 marks the start of Twinkle tagging.